

Exercise for chapter 11

Ke Hua (花珂)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*ke.hua@nwpu.edu.cn

<http://teacher.nwpu.edu.cn/kehua>

Chapter 11

11.1 A gas mixture of 50% CO , 25% CO₂, and 25% H₂ (by volume) is fed to a furnace at 900°C. Determine the composition of the equilibrium CO–CO₂–H₂–H₂O gas if the total pressure of the gas in the furnace is 1 atm.

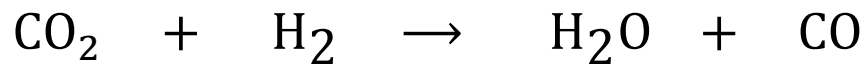
Chapter 11

Solution to 11.1

For $\text{CO}_2 + \text{H}_2 = \text{H}_2\text{O} + \text{CO} \quad \Delta G_{1173}^0 = -1416 \text{ J}$

$$\Delta G^\circ = -RT \ln K_p$$

So $K_p = \exp\left(\frac{1416}{8.3144 \times 1173}\right) = 1.156$



initially, $0.25 \quad 0.25 \quad 0 \quad 0.5$

upon reaction, $0.25 - x \quad 0.25 - x \quad x \quad 0.5 + x$

$$n_T = 0.25 - x + 0.25 - x + x + 0.5 + x = 1$$

Chapter 11

Solution to 11.1

with $P_T = 1 \text{ atm}$

$$K_P = \frac{P_{H_2O} \cdot P_{CO}}{P_{CO_2} \cdot P_{H_2}}$$

$$1.156 = \frac{x(0.5 + x)}{(0.25 - x)^2}$$

$$x = 0.0676$$

So

$$P_{CO_2} = P_{H_2} = 0.25 - 0.0676 = 0.182 \text{ atm}$$

$$P_{CO} = 0.5 + 0.0676 = 0.568 \text{ atm}$$

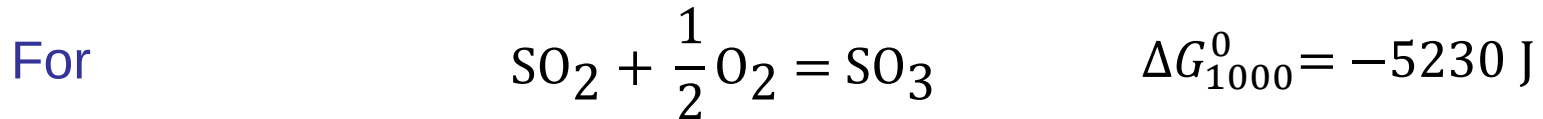
$$P_{H_2O} = 0.0676 \text{ atm}$$

Chapter 11

11. 2 How much heat is evolved when 1 mole of SO_2 and $\frac{1}{2}$ mole of O_2 , each at 1 atm pressure, react to form the equilibrium SO_3 – SO_2 – O_2 mixture at 1000 K and 1 atm pressure?

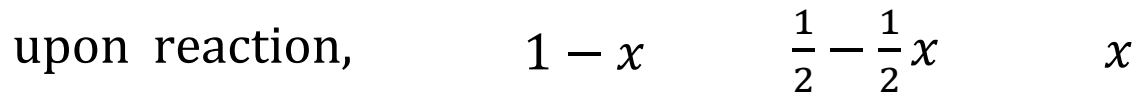
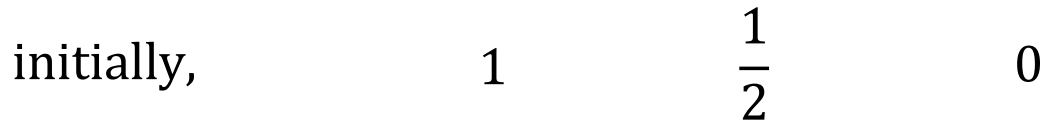
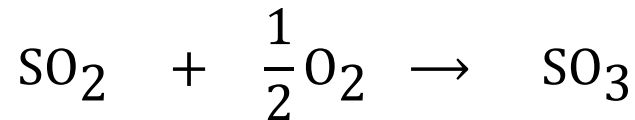
Chapter 11

Solution to 11. 2



$$\Delta G^\circ = -RT \ln K_p$$

So $K_{P,1000} = \exp\left(\frac{5230}{8.3144 \times 1000}\right) = 1.876$



$$n_T = 1 - x + \frac{1}{2} - \frac{1}{2}x + x = \frac{1}{2}(3 - x)$$

Chapter 11

Solution to 11. 2

$$\text{So } P_{SO_2} = \frac{2(1-x)}{3-x} P, \quad P_{O_2} = \frac{1-x}{3-x} P, \quad P_{SO_3} = \frac{2x}{3-x} P$$

$$K_P = \frac{P_{SO_3}}{P_{SO_2} \cdot P_{O_2}^{1/2}} = 1.876 = \frac{x \cdot (3-x)^{1/2}}{(1-x)^{1/2} \cdot (1-x)}$$

$$x = 0.463$$

For formation of 1 mole of SO_3 , $\Delta H = \Delta H^0 = -94600 \text{ J}$

$$\begin{aligned} \text{For formation of 0.463 mole of } SO_3, \quad \Delta H &= 0.463 \times (-94600) \\ &= -43800 \text{ J} \end{aligned}$$

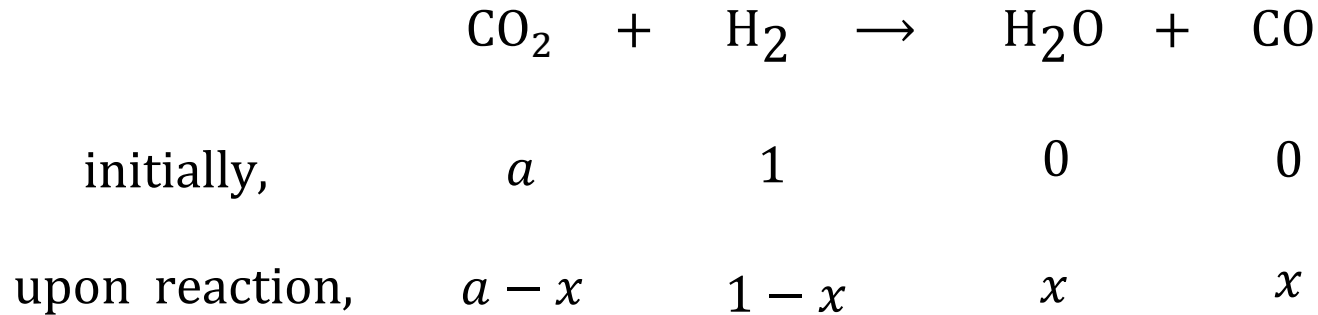
So 43800 J are evolved.

Chapter 11

11. 3 A $\text{CO}_2\text{--CO--H}_2\text{O--H}_2$ gas mixture at a total pressure of 1 atm exerts a partial pressure of oxygen of 10^{-7} atm at 1600°C . In what ratio were the CO_2 and H_2 mixed to produce the gas with this oxygen pressure?

Chapter 11

Solution to 11.3



For $\text{H}_2 + \frac{1}{2}\text{O}_2 = \text{H}_2\text{O} \quad \Delta G_{1873}^0 = -142892 \text{ J}$

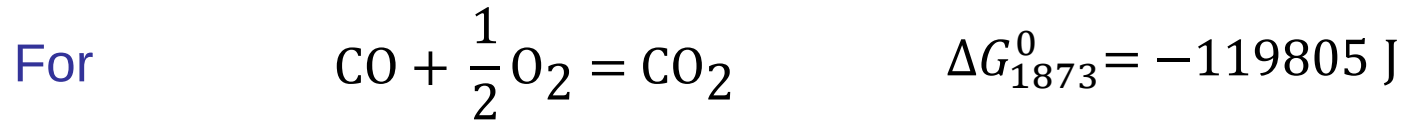
$$\Delta G^\circ = -RT \ln K_p$$

So
$$K_P = \exp\left(\frac{142892}{8.3144 \times 1873}\right) = 9659 = \frac{P_{\text{H}_2\text{O}}}{P_{\text{H}_2} \cdot P_{\text{O}_2}^{1/2}} = \frac{x}{(1-x)10^{-3.5}}$$

$$x = 0.753$$

Chapter 11

Solution to 11.3



So
$$K_P = 2193 = \frac{P_{\text{CO}_2}}{P_{\text{CO}} \cdot P_{\text{O}_2}^{1/2}} = \frac{a - x}{x \cdot 10^{-3.5}} = \frac{a - 0.753}{0.753 \cdot 10^{-3.5}}$$

$$a = 1.276$$

So
$$\frac{\text{CO}_2}{\text{H}_2} = a = 1.276$$

Chapter 11

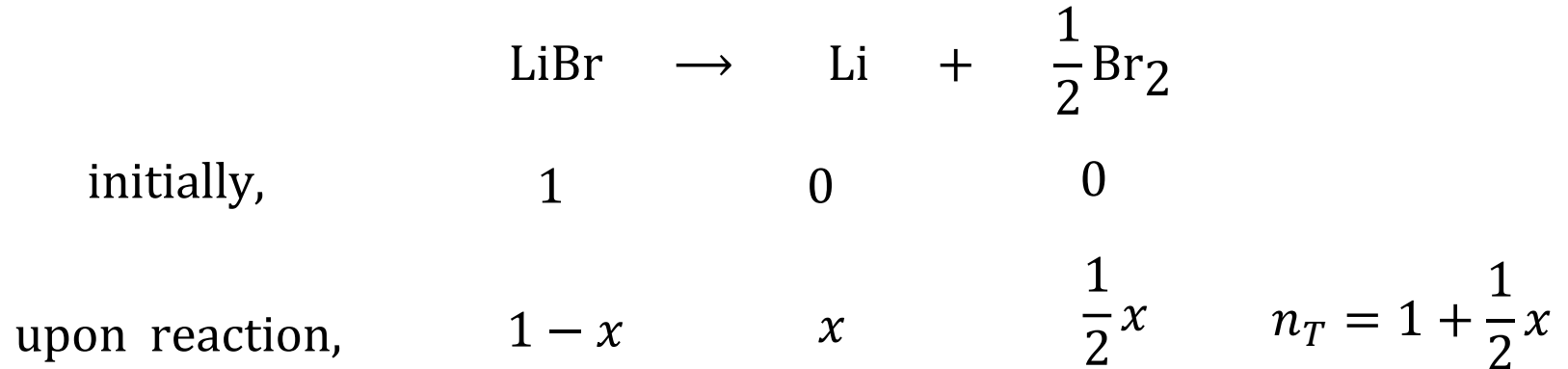
11.4 Lithium bromide vapor dissociates according to



At what temperature does the partial pressure of Li reach the value of 10^{-5} atm when the gas is heated at a constant total pressure of 1 atm?

Chapter 11

Solution to 11.4



So with $P_T = 1 \text{ atm}$, $P_{\text{Li}} = \frac{x}{1 + \frac{1}{2}x} = 10^{-5}$

$$x = 10^{-5}$$

$$P_{\text{LiBr}} = \frac{1-x}{1 + \frac{1}{2}x} \approx 1 \text{ atm}$$

$$P_{\text{Br}_2} = 0.5 \times 10^{-5} \text{ atm}$$

Chapter 11

Solution to 11.4

So

$$K_p = \frac{P_{Li} \cdot P_{Br_2}^{1/2}}{P_{LiBr}} = (0.5 \times 10^{-5})^{1/2} \times 10^{-5}$$

$$= 2.236 \times 10^{-8}$$

$$\Delta G^\circ = -RT \ln K_p$$

$$\Delta G^\circ = 333900 - 42.09 T = -8.3144 T \ln 2.236 \times 10^{-8}$$

$$T = 1771 \text{ K}$$

Chapter 11

11. 5 When SO_3 is decomposed at the constant pressure P and $T = 1000 \text{ K}$, the partial pressure of O_2 in the equilibrium gas is 0.05 atm . What is the pressure P ? If the pressure of this equilibrated gas is increased to 1 atm , to what value must the temperature be decreased to produce a gas mixture in which $P_{\text{O}_2} = 0.05 \text{ atm}$?

Chapter 11

Solution to 11.5

For $\text{SO}_3 = \text{SO}_2 + \frac{1}{2}\text{O}_2 \quad \Delta G_{1000}^{\circ} = 5236 \text{ J}$

So $K_P = 0.533 \quad \Delta G^{\circ} = -RT \ln K_P$

$$P_{\text{SO}_2} = 2P_{\text{O}_2}$$

$$P_{\text{SO}_3} = P_T - 3P_{\text{O}_2}$$

So
$$K_P = 0.533 = \frac{P_{\text{SO}_2} \cdot P_{\text{O}_2}^{1/2}}{P_{\text{SO}_3}} = \frac{2 \times 0.05 \times (0.05)^{1/2}}{(P_T - 0.15)}$$

$$P_T = 0.192 \text{ atm}$$

Chapter 11

Solution to 11.5

$$\text{with } P_T = 1 \text{ atm} \quad K_P = \frac{P_{SO_2} \cdot P_{O_2}^{1/2}}{P_{SO_3}} = \frac{2 \times 0.05 \times (0.05)^{1/2}}{1 - 0.15}$$
$$= 0.0263$$

$$\Delta G^\circ = -RT \ln K_p$$

$$\Delta G^\circ = 94600 - 89.37 T = -8.3144 T \ln 0.0263$$

$$T = 792 \text{ K}$$

Chapter 11

11.6 For the dissociation of nitrogen according to



$$\Delta G^0 = 945,000 - 114.9T \text{ J}$$

- Calculate the equilibrium partial pressure of N in nitrogen gas at 3000 K and a total pressure of 1 atm.
- The total pressure of the gas, at 3000 K, at which the partial pressure of N_2 is 90% of the total pressure.

Chapter 11

Solution to 11.6

a. For $N_2 = 2N$ $\Delta G_{3000}^0 = 600300 \text{ J}$

So $K_P = 3.53 \times 10^{-11} = \frac{P_N^2}{(1 - P_N)}$ $\Delta G^\circ = -RT \ln K_p$

$$P_N = 5.94 \times 10^{-6} \text{ atm}$$

b. For $P_T = P_{N_2} + P_N$

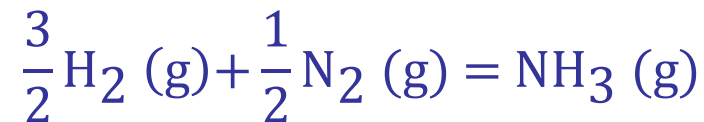
$$= 0.9P_T + 0.1P_T$$

So $K_P = 3.53 \times 10^{-11} = \frac{(0.1P_T)^2}{0.9P_T} = \frac{0.01}{0.9} P_T$

$$P_T = 3.18 \times 10^{-9} \text{ atm}$$

Chapter 11

11.7 Ammonia gas is heated to 300°C. At what total pressure is the mole fraction of N₂ in the equilibrium gas mixture equal to 0.2 ? Calculate the standard enthalpy change and the standard entropy change for the reaction



at 300°C.

Chapter 11

Solution to 11.7

For $2\text{NH}_3 = \text{N}_2 + 3\text{H}_2$ $\Delta G_{573}^0 = -25021 \text{ J}$

$$\Delta G^\circ = -RT \ln K_p$$

So $K_P = 190.1$

As $P_{\text{H}_2} = 3P_{\text{N}_2}$ $x_{\text{H}_2} = 3x_{\text{N}_2}$

So $x_{\text{N}_2} = 0.2$, $x_{\text{H}_2} = 0.6$, $x_{\text{NH}_3} = 1 - 0.2 - 0.6 = 0.2$

So
$$K_P = 190.1 = \frac{P_{\text{N}_2} \cdot P_{\text{H}_2}^3}{P_{\text{NH}_3}^2} = \frac{x_{\text{N}_2} \cdot x_{\text{H}_2}^3 \cdot P_T^2}{x_{\text{NH}_3}^2} = \frac{0.2 \times 0.6^3 \times P_T^2}{0.2^2}$$

$$P_T = 13.3 \text{ atm}$$

Chapter 11

Solution to 11.7

$$\Delta G^0 = -87030 + 25.8 T \ln T + 31.7T$$



So
$$\frac{\partial \Delta G^0}{\partial T} = -\Delta S^0 = 25.8 \ln T + 25.8 + 31.7$$

So
$$\Delta S_{573}^0 = -221.4 \text{ J/K} \quad \Delta H_{573}^0 = \Delta G_{573}^0 + 573 \cdot \Delta S_{573}^0$$
$$= -25021 + 221.4 \times 573$$

For
$$\frac{3}{2}\text{H}_2 (\text{g}) + \frac{1}{2}\text{N}_2 (\text{g}) = \text{NH}_3 (\text{g}) \quad = -101841 \text{ J}$$

So
$$\Delta H_{573}^0 = -50900 \text{ J} \quad \text{and} \quad \Delta S_{573}^0 = -110.7 \text{ J/K}$$

Chapter 11

11.8 By establishing the equilibrium



at 500 K in a mixture of PCl_5 and PCl_3 , a gas is obtained at 1 atm total pressure in which the partial pressure of Cl_2 is 0.1 atm. In what ratio were PCl_5 and PCl_3 mixed to obtain this equilibrium gas?

Chapter 11

Solution to 11.8

For $\text{PCl}_3 + \text{Cl}_2 = \text{PCl}_5$ $\Delta G_{500}^0 = -2672 \text{ J}$

$$\Delta G^0 = -RT \ln K_p$$

So $K_p = 1.901$



initially, $a \quad 0 \quad 1$

upon reaction, $a + x \quad x \quad 1 - x \quad n_T = a + 1 + x$

with $P_T = 1 \text{ atm}$ $P_{\text{Cl}_2} = \frac{x}{a + 1 + x} = 0.1$

Chapter 11

Solution to 11.8

$$K_p = 1.901 = \frac{P_{\text{PCl}_5}}{P_{\text{PCl}_3} \cdot P_{\text{Cl}_2}} = \frac{(1-x)(a+1+x)}{x \cdot (a+x)}$$

So $a = 2.692, \quad x = 0.410$

$$\frac{P_{\text{Cl}_3}}{P_{\text{Cl}_5}} = a = 2.692$$

Chapter 11

11.9 Air and hydrogen are mixed in the ratio 1/4 and are heated to 1200 K. Calculate the partial pressures of H_2 and O_2 in the equilibrium gas at 1 atm pressure and at 10 atm pressure. Air contains 21 volume percent O_2 .

Chapter 11

Solution to 11.9

$$n_{H_2} = 20 \qquad n_{air} = 80$$

So $n_{O_2} = 16.8 \qquad n_{H_2} = 63.2$

For $H_2 + \frac{1}{2}O_2 = H_2O \qquad \Delta G_{1200}^0 = -180480 \text{ J}$

$$\Delta G^\circ = -RT \ln K_p$$

So
$$K_P = 7.178 \times 10^7 = \frac{P_{H_2O}}{P_{H_2} \cdot P_{O_2}^{1/2}}$$

$$\frac{n_{H_2}}{n_{O_2}} = \frac{20}{16.8} = 1.1905 = \frac{2n_{H_2} + 2n_{H_2O}}{2n_{O_2} + n_{H_2O}}$$

Chapter 11

Solution to 11.9

$$\frac{n_{O_2}}{n_{H_2}} = \frac{16.8}{63.2} = 0.2658 = \frac{2n_{O_2} + 2n_{H_2O}}{126.4}$$

So

$$2n_{O_2} + n_{H_2O} = 0.2658 \times 126.4 = 33.6$$

$$2n_{H_2} + 2n_{H_2O} = 33.6 \times 1.1905 = 40$$

So

$$n_{H_2O} = 20 - n_{H_2} = 33.6 - 2n_{O_2} \quad n_{O_2} = 6.8 + \frac{1}{2}n_{H_2}$$

$$n_T = 20 - n_{H_2} + n_{H_2} + 6.08 + \frac{1}{2}n_{H_2} + 63.2$$

$$= 90 + \frac{1}{2}n_{H_2}$$

Chapter 11

Solution to 11.9

$$K_P = 7.178 \times 10^7 = \frac{n_{H_2O} \cdot n_T^{1/2}}{n_{H_2} \cdot n_{O_2}^{1/2} \cdot P^{1/2}}$$

$$\text{with } P_T = 1 \text{ atm, } 7.178 \times 10^7 = \frac{(20 - n_{H_2}) \cdot \left(90 + \frac{1}{2}n_{H_2}\right)^{1/2}}{n_{H_2} \cdot \left(6.8 + \frac{1}{2}n_{H_2}\right)^{1/2}}$$

So $n_{H_2} = 9.43 \times 10^{-7} \text{ moles}$ $n_{O_2} = 6.8 \text{ moles}$ $n_T = 90 \text{ moles}$

So $P_{H_2} = \frac{n_{H_2}}{n_T} \cdot P_T = \frac{9.43 \times 10^{-7}}{90} \times 1 = 1.05 \times 10^{-8} \text{ atm}$

$$P_{O_2} = \frac{6.8}{90} \times 1 = 0.0756 \text{ atm}$$

Chapter 11

Solution to 11.9

$$\text{with } P_T = 10 \text{ atm,} \quad 7.178 \times 10^7 = \frac{(20 - n_{H_2}) \cdot \left(90 + \frac{1}{2}n_{H_2}\right)^{1/2}}{n_{H_2} \cdot \left(6.8 + \frac{1}{2}n_{H_2}\right)^{1/2} \cdot \sqrt{10}}$$

$$\text{So} \quad n_{H_2} = 2.98 \times 10^{-7} \text{ moles} \quad n_{O_2} = 6.8 \text{ moles}$$

$$\text{So} \quad P_{H_2} = \frac{n_{H_2}}{n_T} \cdot P_T = \frac{2.98 \times 10^{-7}}{90} \times 10 = 3.31 \times 10^{-8} \text{ atm}$$

$$P_{O_2} = \frac{6.8}{90} \times 10 = 0.756 \text{ atm}$$

Chapter 11

11.10 One mole of each of hydrogen, iodine vapor, and HI gas are allowed to react at 1500 K and $P = 1$ atm. Calculate the mole fractions of H_2 , I_2 , and HI in the equilibrium mixture. The temperature is then changed to that value at which P_{HI} in the equilibrated gas is five times P_{H_2} . What is this temperature?

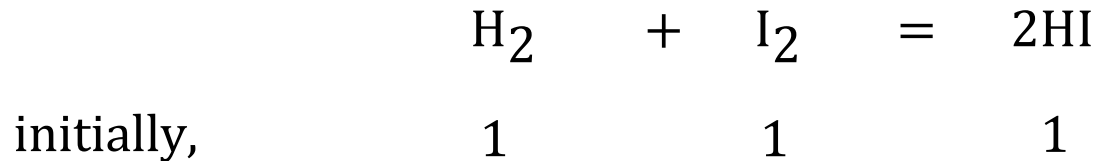
Chapter 11

Solution to 11.10

For $\text{H}_2 + \text{I}_2 = 2\text{HI}$ $\Delta G_{1500}^0 = -34845 \text{ J}$

$$\Delta G^0 = -RT \ln K_p$$

So $K_P = 16.345 = \frac{P_{\text{HI}}^2}{P_{\text{H}_2} \cdot P_{\text{I}_2}}$



upon reaction,	$1 - x$		$1 - x$		$1 + 2x$	$n_T = 3$
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So $K_P = \frac{(1 + 2x)^2}{(1 - x)^2} = 16.345$ $x = 0.504$

Chapter 11

Solution to 11.10

So $x_{H_2} = x_{I_2} = \frac{1 - 0.504}{3} = 0.165,$

$$x_{HI} = \frac{1 + 2 \times 0.504}{3} = 0.669$$

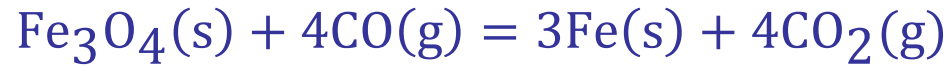
For $P_{HI} = 5P_{H_2}$ $K_P = \frac{(5P_{H_2})^2}{P_{HI}^2} = 25$

So $\Delta G^0 = -8370 - 17.65T = -8.3144 T \ln 25$

$$T = 918 \text{ K}$$

Chapter 11

11.11 Consider the reduction of magnetite by CO to form pure iron :



Calculate the partial pressure of CO and CO₂ in terms of the equilibrium constant K_P for the reaction.

Chapter 11

Solution to 11.11

Assume the activity of the solids to be unity and that the gases behave as ideal gases.

For
$$K_P = \frac{a_{Fe}^3 \cdot a_{CO_2}^4}{a_{Fe_3O_4} \cdot a_{CO}^4} = \frac{P_{CO_2}^4}{P_{CO}^4}$$

So
$$K_P^{1/4} = \frac{P_{CO_2}}{P_{CO}}$$

Also the total pressure is 1 atmosphere : $P_{CO_2} + P_{CO} = 1$

Solving the two equations simultaneously:

$$P_{CO} = \frac{1}{\left(K_P^{1/4} + 1\right)} \qquad P_{CO_2} = \frac{K_P^{1/4}}{\left(K_P^{1/4} + 1\right)}$$

Chapter 11

11.12 It is known that at 300 K, the value of K_p for a certain reaction is 10^{12} . For the reaction, ΔH° is 100 kJ/mole.

a. Determine if this reaction is favorable at 800 K and estimate K_p (800K). Explain.

b. The actual value of K_p (800) is 35. Explain any discrepancy from your estimate.

Chapter 11

Solution to 11.12

a. First, estimate K_p at 800 K.

$$\ln \left(\frac{K_p(800\text{K})}{K_p(300\text{K})} \right) = \frac{\Delta H^0}{R} \left(\frac{1}{300} - \frac{1}{800} \right)$$

$$\ln \left(\frac{K_p(800\text{K})}{K_p(300\text{K})} \right) = \frac{-100000}{8.3144} \left(\frac{1}{300} - \frac{1}{800} \right) = -25.06$$

$$\frac{K_p(800\text{K})}{K_p(300\text{K})} = \exp(-25.06)$$

$$K_p(800\text{K}) = K_p(300\text{K}) \cdot \exp(-25.06) = 13.12$$

Since the value is greater than unity the reaction is possible at 800 K.

b. The above method assumed there was no temperature dependence of ΔH^0 over the span of 500 K.

Chapter 11

11.13 Sketch the natural log of K_p versus $1/T$ for the following cases:

a. $\Delta H^0 < 0, \Delta S^0 < 0$

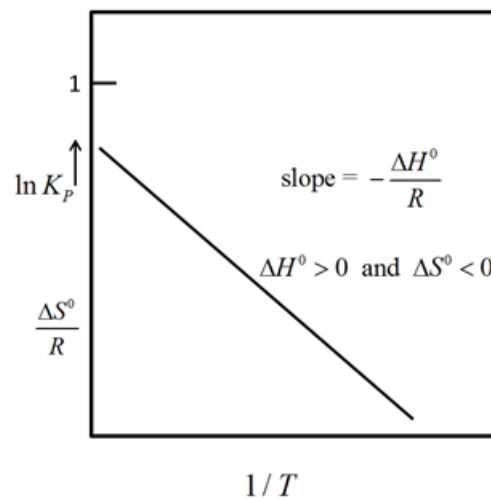
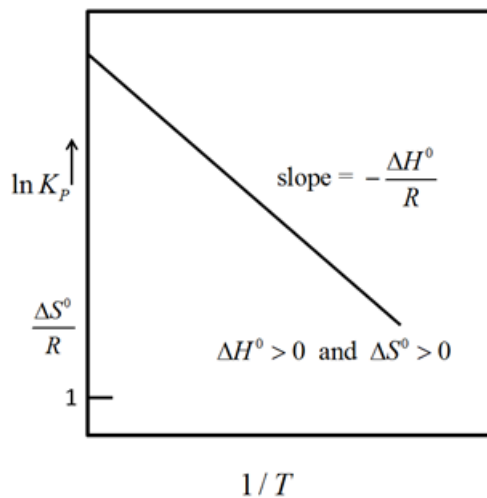
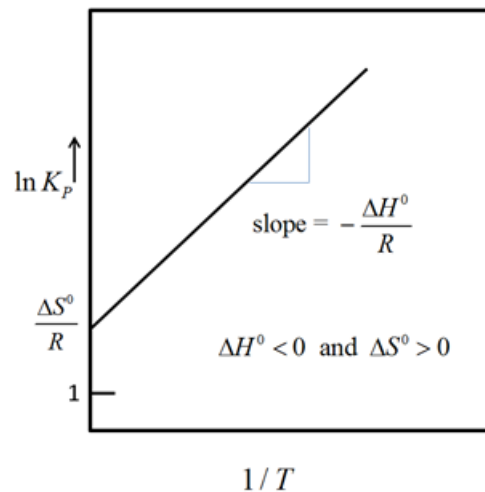
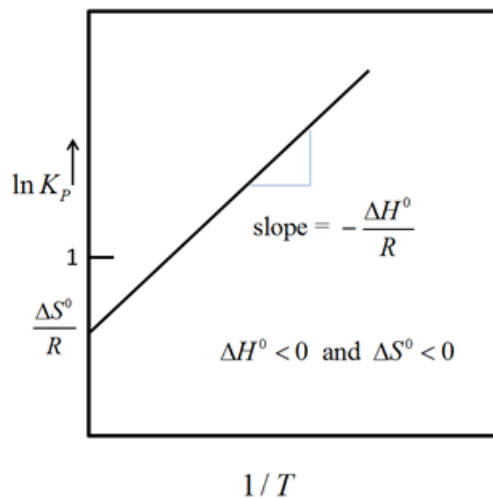
b. $\Delta H^0 < 0, \Delta S^0 > 0$

c. $\Delta H^0 > 0, \Delta S^0 > 0$

d. $\Delta H^0 > 0, \Delta S^0 < 0$

Chapter 11

Solution to 11.13



Thermodynamics of Materials & Phase transformations

The End

